

ROCKWELL CONTROLLOGIX 5580 (1756-L950TPSXT) — 5-CHASSIS DLR NETWORK CABLE & 13-SLOT REDESIGN

DRAWING NO.
DWG-260911-NET-01

REVISION
REV B02

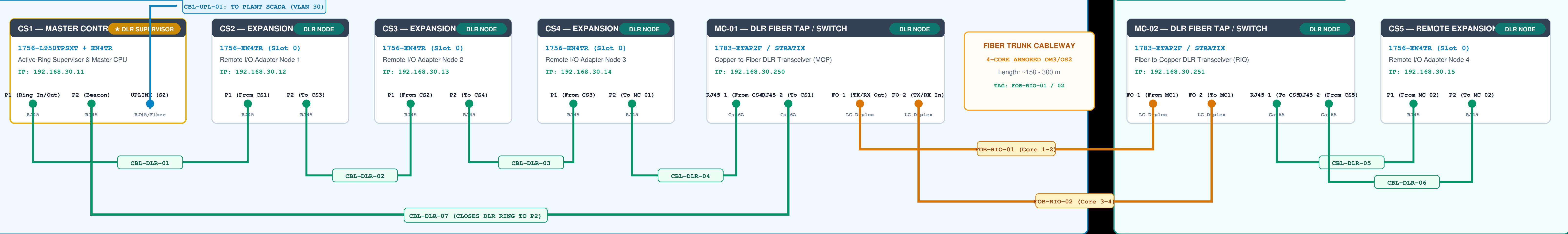
TOPOLOGY

ODVA DLR RING

CONTROLLER MODEL
1756-L950TPSXT (XT)

CHASSIS SPEC
5 CHASSIS x 13 SLOTS

MAIN CONTROL PANEL (MCP-01) — SAME ENCLOSURE (CS1 - CS4)



STANDARDIZED BALANCED I/O ARCHITECTURE (4 DI / 2 DO / 2 AI / 2 AO / 2 SPARE PER RACK)

CHASSIS 1 — MASTER CONTROLLER (MCP-01)

Location: Main Control Panel / Rack 1

Network: Active DLR Supervisor + Supervisory Master

1756-A13 (13)

PWR

1756

PA72

DC OK

SLOT 00

CPU

CPU

1756-L950TPSXT

GuardLogix Process Safety Controller

SLOT 01

Eth_Module

1756-EN4TR

DLR Ring Supervisor (1Gbps Dual Port)

SLOT 02

Eth_Module

1756-EN4TR

Plant SCADA / FactoryTalk Uplink

SLOT 03

32DI

1756-IB32

24VDC Sink/Source Digital Input (DI-)

SLOT 04

32DI

1756-IB32

24VDC Sink/Source Digital Input (DI-)

SLOT 05

32DI

1756-IB32

24VDC Sink/Source Digital Input (DI-)

SLOT 06

32DI

1756-IB32

24VDC Sink/Source Digital Input (DI-)

SLOT 07

32DO

1756-OB32

24VDC Sourcing Digital Output (DO-01)

SLOT 08

32DO

1756-OB32

24VDC Sourcing Digital Output (DO-02)

SLOT 09

16AI

1756-IF16

16-Ch High-Speed Analog Input (AI-01)

SLOT 10

16AI

1756-IF16

16-Ch High-Speed Analog Input (AI-02)

SLOT 11

8AO

1756-OF8

8-Ch Voltage/Current Analog Output (AO-01)

SLOT 12

8AO

1756-OF8

8-Ch Voltage/Current Analog Output (AO-02)

CHASSIS 2 — EXPANSION RACK 1 (MCP-01)

Location: Main Control Panel / Rack 2

Network: DLR Ring Node 1

1756-A13 (13)

PWR

1756

PA72

DC OK

SLOT 00

Eth_Module

1756-EN4TR

Remote I/O Adapter (Slot 0 Redesign)

SLOT 01

32DI

1756-IB32

24VDC Digital Input (DI-05)

SLOT 02

32DI

1756-IB32

24VDC Digital Input (DI-06)

SLOT 03

32DI

1756-IB32

24VDC Digital Input (DI-07)

SLOT 04

32DI

1756-IB32

24VDC Digital Input (DI-08)

SLOT 05

32DO

1756-OB32

24VDC Digital Output (DO-03)

SLOT 06

32DO

1756-OB32

24VDC Digital Output (DO-04)

SLOT 07

16AI

1756-IF16

16-Ch Analog Input (AI-03)

SLOT 08

16AI

1756-IF16

16-Ch Analog Input (AI-04)

SLOT 09

8AO

1756-OF8

8-Ch Analog Output (AO-03)

SLOT 10

8AO

1756-OF8

8-Ch Analog Output (AO-04)

SLOT 11

Empty

1756-N2

Slot Filler / Spare Expansion

SLOT 12

Empty

1756-N2

Slot Filler / Spare Expansion

CHASSIS 3 — EXPANSION RACK 2 (MCP-01)

Location: Main Control Panel / Rack 3

Network: DLR Ring Node 2

1756-A13 (13)

PWR

1756

PA72

DC OK

SLOT 00

Eth_Module

1756-EN4TR

Remote I/O Adapter (Slot 0 Redesign)

SLOT 01

32DI

1756-IB32

24VDC Digital Input (DI-09)

SLOT 02

32DI

1756-IB32

24VDC Digital Input (DI-10)

SLOT 03

32DI

1756-IB32

24VDC Digital Input (DI-11)

SLOT 04

32DI

1756-IB32

24VDC Digital Input (DI-12)

SLOT 05

32DO

1756-OB32

24VDC Digital Output (DO-05)

SLOT 06

32DO

1756-OB32

24VDC Digital Output (DO-06)

SLOT 07

16AI

1756-IF16

16-Ch Analog Input (AI-05)

SLOT 08

16AI

1756-IF16

16-Ch Analog Input (AI-06)

SLOT 09

8AO

1756-OF8

8-Ch Analog Output (AO-05)

SLOT 10

8AO

1756-OF8

8-Ch Analog Output (AO-06)

SLOT 11

Empty

1756-N2

Slot Filler / Spare Expansion

SLOT 12

Empty

1756-N2

Slot Filler / Spare Expansion

CHASSIS 4 — EXPANSION RACK 3 (MCP-01)

Location: Main Control Panel / Rack 4

Network: DLR Ring Node 3

1756-A13 (13)

PWR

1756

PA72

DC OK

SLOT 00

Eth_Module

1756-EN4TR

Remote I/O Adapter (Slot 0 Redesign)

SLOT 01

32DI

1756-IB32

24VDC Digital Input (DI-13)

SLOT 02

32DI

1756-IB32

24VDC Digital Input (DI-14)

SLOT 03

32DI

1756-IB32

24VDC Digital Input (DI-15)

SLOT 04

32DI

1756-IB32

24VDC Digital Input (DI-16)

SLOT 05

32DO

1756-OB32

24VDC Digital Output (DO-07)

SLOT 06

32DO

1756-OB32

24VDC Digital Output (DO-08)

SLOT 07

16AI

1756-IF16

16-Ch Analog Input (AI-07)

SLOT 08

16AI

1756-IF16

16-Ch Analog Input (AI-08)

SLOT 09

8AO

1756-OF8

8-Ch Analog Output (AO-07)

SLOT 10

8AO

1756-OF8

8-Ch Analog Output (AO-08)

SLOT 11

Empty

1756-N2

Slot Filler / Spare Expansion

SLOT 12

Empty

1756-N2

Slot Filler / Spare Expansion

CHASSIS 5 — REMOTE I/O RACK (RIO200 PANEL)

Location: Remote I/O Panel RIO200 (Area 200) via Fiber

Network: DLR Ring Node 4 (Fiber Link)

1756-A13 (13)

PWR

1756

PA72

DC OK

SLOT 00

Eth_Module

1756-EN4TR

Remote I/O Adapter (Slot 0 Redesign)

SLOT 01

32DI

1756-IB32

24VDC Digital Input (DI-17)

SLOT 02

32DI

1756-IB32

24VDC Digital Input (DI-18)

SLOT 03

32DI

1756-IB32

24VDC Digital Input (DI-19)

SLOT 04

32DI

1756-IB32

24VDC Digital Input (DI-20)

SLOT 05

32DO

1756-OB32

24VDC Digital Output (DO-09)

SLOT 06

32DO

1756-OB32

24VDC Digital Output (DO-10)

SLOT 07

16AI

1756-IF16

16-Ch Analog Input (AI-09)

SLOT 08

16AI

1756-IF16

16-Ch Analog Input (AI-10)

SLOT 09

8AO

1756-OF8

8-Ch Analog Output (AO-09)

SLOT 10

8AO

1756-OF8

8-Ch Analog Output (AO-10)

SLOT 11

Empty

1756-N2

Slot Filler / Spare Expansion

SLOT 12

Empty

1756-N2

Slot Filler / Spare Expansion

CARD CODE	MODEL NO.	DESCRIPTION	Q'TY	CHANNELS
CPU	1756-L950TPSX	ControlLogix 5580 Process Safety (XT)	1	50MB Memory
Eth_Module	1756-EN4TR	Dual-Port 1Gbps DLR EtherNet/IP Adapter	6	DLR / Uplink
32DI	1756-IB32	24VDC Digital Input (Sink/Source)	20	640 DI pts
32DO	1756-OB32	24VDC Sourcing Digital Output	10	320 DO pts
16AI	1756-IF16	16-Ch Voltage/Current Analog Input	10	160 AI pts
8AO	1756-OF8	8-Ch Voltage/Current Analog Output	10	80 AO pts
Empty	1756-N2	Chassis Slot Filler Covers (Spares)	8	Future Exp
TOTAL SLOTS ALLOCATED		5 CHASSIS x 13 SLOTS = 65 SLOTS	65	1,200 I/O

Cable Tag	From (Port)	To (Port)	Media / Spec	Est. Len
CBL-DLR-01	CS1-EN4TR (P1)	CS2-EN4TR (P1)	Cat6A Shielded S/FTP	1.5 m
CBL-DLR-02	CS2-EN4TR (P2)	CS3-EN4TR (P1)	Cat6A Shielded S/FTP	1.5 m
CBL-DLR-03	CS3-EN4TR (P2)	CS4-EN4TR (P1)	Cat6A Shielded S/FTP	1.5 m
CBL-DLR-04	CS4-EN4TR (P2)	MC-01 (RJ45-1)	Cat6A Shielded S/FTP	2.0 m
FOB-RIO-01	MC-01 (FO-1 TX/RX)	MC-02 (FO-1 TX/RX)	4C Armored OM3/OS2	200 m
CBL-DLR-05	MC-02 (RJ45-1)	CS5-EN4TR (P1)	Cat6A Shielded S/FTP	1.5 m
CBL-DLR-06	CS5-EN4TR (P2)	MC-02 (RJ45-2)	Cat6A Shielded S/FTP	1.5 m
FOB-RIO-02	MC-02 (FO-2 TX/RX)	MC-01 (FO-2 TX/RX)	4C Armored OM3/OS2	200 m
CBL-DLR-07	MC-01 (RJ45-2)	CS1-EN4TR (P2 Beacon)	Cat6A S/FTP (Ring Close)	2.0 m
CBL-UPL-01	CS1-EN4TR (S02)	Core Switch (VLAN 30)	Cat6A / 1Gbps Fiber	15 m

1. CS1 Slot 1 is the ACTIVE RING SUPERVISOR. Port 2 is beacon-blocked during normal operation.
2. DLR Beacon Interval configured at 400µs; single cable fault failover recovers in < 3ms.
3. Expansion chassis CS2–CS5 standardized with 1756-EN4TR in Slot 00 per Rockwell CPwE guideline